# EXERCISE

1. Which of the following are functions ? If they are functions, then find their range.

(1) $f : R \to R, \quad f(x) = 2x - 1.$

(2) $g : R \to R, \quad g(x) = x, \ x \geq 0$

$= -x, \ x < 0$

(3) $h : R^+ \to R, \ h(x) = \sqrt{x}.$

(4) $f : R \to R, \ f(x) = 3x + 4.$

(5) $g : R \to R, \ g(x) = |x|.$

(6) $A = \{1, 2, 3\}, \ f : A \to R, \ f(x) = 2^x.$

(7) $A = \{1, 2, 3\}$, $g : A \to R$, $g(x) = |x - 2|$.

(8) $A = \{0, 1, 2\}$, $f : A \to R$, $f(x) = x^2 - 3x + 2$.

**Ans. :** (1) is a *function*, $R_f = R$.

(2) is a *function*, $R_g = R^+ \cup \{0\}$.

(3) not a *function*.

(4) is a *function*, $R_f = R$.

(5) is a *function*, $R_g = R^+ \cup \{0\}$.

(6) is a *function*, $R_f = \{2, 4, 8\}$.

(7) is a *function*, $R_g = \{0, 1\}$.

(8) is a *function*, $R_f = \{0, 2\}$.

**2.** $f : R - \{-1\} \to R$, $f(x) = \dfrac{1 - x}{1 + x}$ then prove that

(i) $f(x) \cdot f(y) = f\left(\dfrac{x + y}{1 + xy}\right)$.

(ii) $f(x) + f\left(\dfrac{1}{x}\right) = 0$, $(x \neq 0)$.

(iii) $f(f(x)) = x$.

(iv) $f(x) - f\left(\dfrac{1}{x}\right) = 2f(x)$, $x \neq 0$.

**3.** Which of the following functions are one - one ? Is it onto ?

(1) $f : R \to R$, $f(x) = 2x - 1$. **Ans. :** one-one, onto

(2) $g : N \to Z$, $g(n) = n - 1$. **Ans. :** one-one, not onto.

(3) $h : R \to R$, $h(x) = x^2$. **Ans. :** not one-one, not onto.

(4) $f : N \to N, f(x) = 2x - 1$. **Ans. :** one-one, not onto.

(5) $g : R \to R, g(x) = 3x + 5$. **Ans. :** one-one, onto.

(6) $h : R \to R, h(x) = |x|$. **Ans. :** not one-one, not onto.

**4.** $A = \{0, 2, 4\}, B = \{1, 3, 5\}, f : A \to B,$

$f = \{(0, 1), (2, 3), (4, 5)\}$ and $g : B \to A,$

$g : \{(1, 0), (3, 2), (5, 4)\}$ then find $gof$ and $fog$.

**Ans. :** $gof = \{(0, 0), (2, 2), (4, 4)\}$

$fog = \{(1, 1), (3, 3), (5, 5)\}$.

**5.** $f : R \to R, f(x) = x - 1$ and $g : R \to R, g(x) = x + 2$ then prove that $fog = gof$.

**6.** $f : R \to R, f(x) = 2x - 1, g : R \to R, g(x) = 3x + k$ and $gof = fog$, then find $k$.

**Ans. :** $k = -2$.

**7.** Verify whether the inverse of the following functions exists or not. If it exists, find it.

(i) $f : Z \to Z, f(x) = x + 2$.

(ii) $f : N \to N, f(x) = x + 2$.

(iii) $f : R \to R, f(x) = x^2 + 1$.

(iv) $f : R \to R, f(x) = 2x + 5$.

**Ans. :** (i) exists, $f^{-1}(x) = x - 2$.

(ii) does not exist.

(iii) does not exist.

(iv) exists, $f^{-1}(x) = \dfrac{x - 5}{2}$.

**8.** $f : N \to N, f(x) = 2x + 1$ and $g : N \to N, g(x) = x + 2$, then find $gof$ and $fog$. Also show that $gof \neq fog$.

**Ans. :** $gof : N \to N, \quad gof(x) = 2x + 3.$

$fog : N \to N, \quad fog(x) = 2x + 5.$

**9.** If $f : R \to R \times R, \; f(x) = (x, x)$ and $g : R \times R \to R$, $g(x, y) = \dfrac{x + y}{2}$ then find $fog$ and $gof$.

**Ans. :** $gof : R \to R, \; g(x) = x$

$fog : R \times R \to R \times R, \; fog(x, y) = \left(\dfrac{x + y}{2}, \dfrac{x + y}{2}\right).$

**10.** If for $f : R \to R$ and $g : R \to R, f(x) = \dfrac{x}{2} + 3$ and $g(x) = 2x - k$ and $gof = fog$ then find $k$.

**Ans. :** 6.

**11.** $f : R \to R, f(x) = \dfrac{x + 1}{2}, g : R \to R, g(x) = 2x - 1$ and $h : R \to R, h(x) = x^2$, then prove that $ho(gof) = (hog)of$.

**12.** $f : R \to R, \; f(x) = 2x, g : R \to R, g(x) = x + 2$ and $h : R \to R, \; h(x) = \dfrac{x}{2}$ then prove that $ho(gof) = (hog)of$.

**13.** $f : R \to R, \; f(x) = x^2, \; g : R \to R, \; g(x) = 3x - 5$ and $h : R \to R, h(x) = 2x$. Then find $fog$, $gof$, $fo(goh)$, $(fog)oh$.

**Ans. :** (i) $fog : R \to R \qquad fog(x) = (3x - 5)^2$

(ii) $gof : R \to R \qquad gof(x) = 3x^2 - 5$

(iii) $fo(goh) : R \to R \qquad fo(goh)(x) = (6x - 5)^2$

(iv) $(fog)oh : R \to R \qquad (fog)oh(x) = (6x - 5)^2.$

**14.** For the exponential function

$$f = \left\{(x, y) \,/\, y = \left(\sqrt{2}\right)^x,\ x \in R\right\} \text{ find } D_f, R_f, f(0), f(8).$$

**Ans. :** $R$, $R^+$, 1, 16.

**15.** If $f(x) = e^x$, $x \in R$ then prove that

$$\frac{f(x+h) - f(x)}{h} = \frac{e^x\left(e^h - 1\right)}{h} \quad (h \neq 0).$$

**16.** Find $x$, if $2^{2+2x} + 4^{1-x} = 10$. **Ans. :** $\pm \frac{1}{2}$.

**17.** Find $x$, if $\dfrac{3^{x-1} + 3^x}{2^x + 2^{x+1}} = \dfrac{3}{2}$. **Ans. :** $x = 3$.

**18.** If $f(x) = log_e\, x$, $x \in R^+$ then prove that

$$f(a) + f(b) = 2f\left(\frac{a+b}{2}\right) \Leftrightarrow a = b.$$

**19.** If $a^x = b^y = ab$, then prove that $\dfrac{1}{x} + \dfrac{1}{y} = 1$,

where $a, b \in R^+ - \{1\}$.

**20.** If $log\left(\dfrac{x-y}{3}\right) = \dfrac{1}{2}(log\, x + log\, y)$, then prove that

$x^2 + y^2 = 11xy$, where $(x > y > 0)$.

**21.** If $f(x) = 2^x$, $x \in R$ and $g(x) = \log_4 x$, $x \in R^+$ then find $(gof)(4)$, $(fog)(8)$. Also find $a$, if $f(0) = g(a)$.

**Ans. :** 2, $2\sqrt{2}$, $a = 4$.

**22.** If $f(x) = log\left(\frac{1+x}{1-x}\right)$, $0 < x < 1$ then prove that

$$f\left(\frac{2x}{1+x^2}\right) = 2\,f(x).$$